Build Web Apps You Can Formally Verify.

A Rust web framework with Lean 4 verification.

900+
GITHUB STARS

80
FORKS

3.8k+
DOWNLOADS

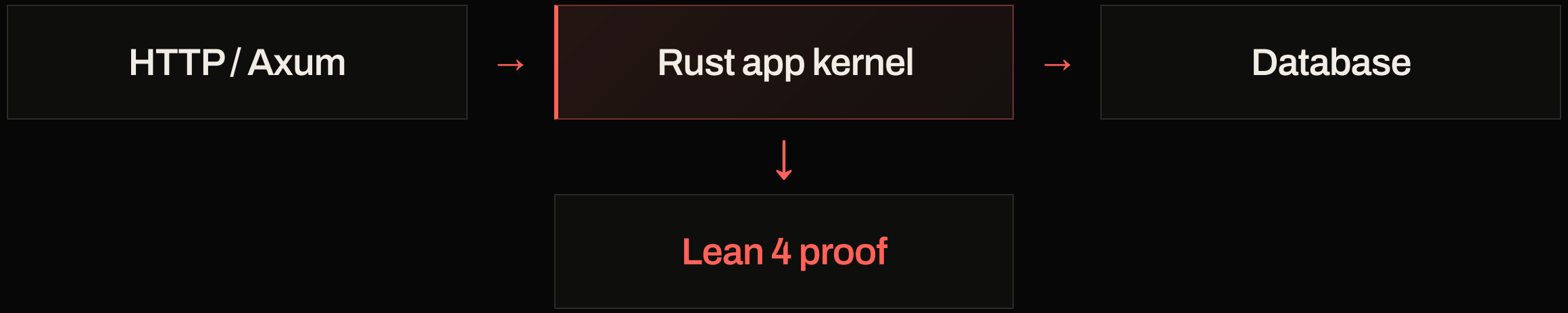
Are You Sure AI-Generated Code Is Correct?

- AI can generate code in seconds.
- But thousands of AI-generated test cases cannot guarantee its correctness.

> → What if correctness could be mathematically proven?

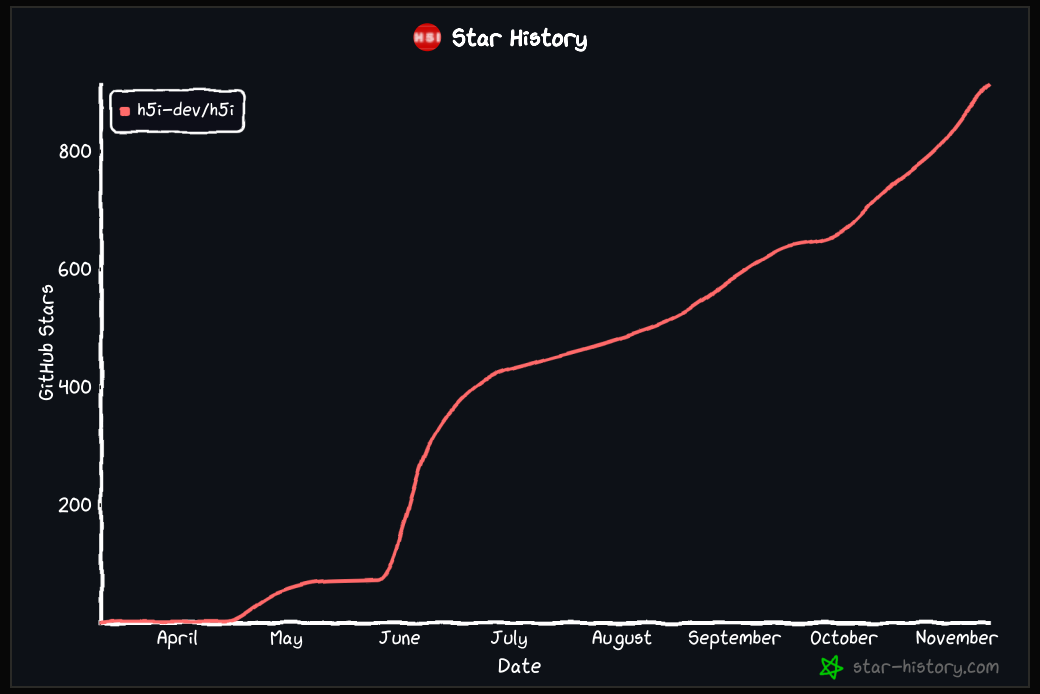
03 / SOLUTION

A Web Framework Built for Formal Verification.



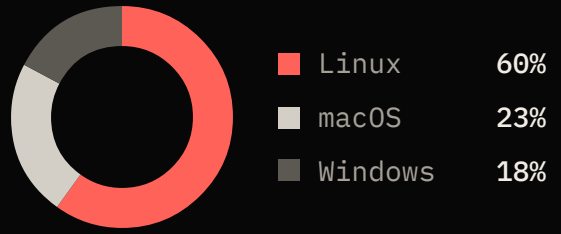
Proofs cover stated properties for all modeled inputs, under explicit assumptions

900+ GitHub Stars and 3.8K+ CLI Downloads



- 80 forks
- 20+ contributors

CLI DOWNLOADS BY OS



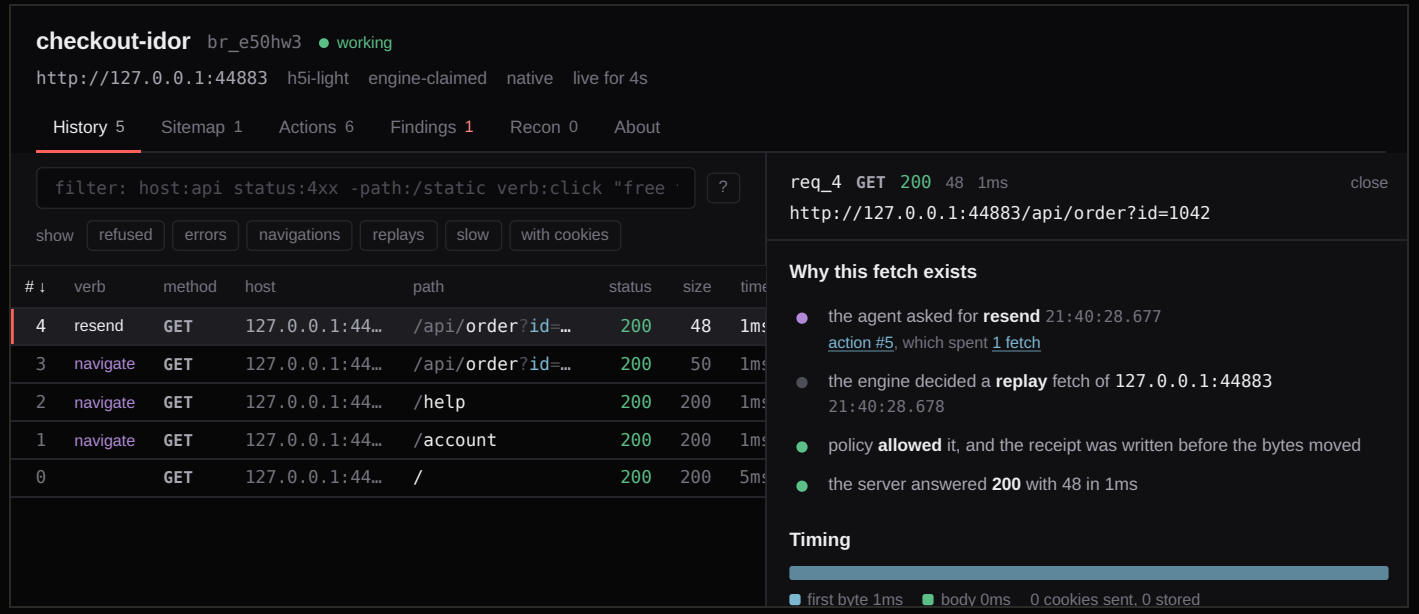
We Plan to Sell Verification Infrastructure on Top of the Open-Source Framework.

FREE · TODAY

- Web framework
- Lean 4 proofs
- Local CLI

PAID · PLANNED

- Continuous verification
- Proof monitoring in CI/CD
- Team collaboration and audit reports



Actual h5i ui dashboard, static capture of a local demo

Hideaki Takahashi

- Columbia CS PhD in software security, formal methods and AI
- Publications at IEEE S&P '26, ACM CCS '26, ICLR '24, AAMAS '24, CVPR '23
- Found 100+ zero-day vulnerabilities and CVEs

> Seeking design partners and seed investors.

Start a conversation ↗ | Explore h5i ↗